

Model Improvement

- Model Prediction gijmatini xaqiqiy gijmatga yaqinlashtirish
 - Regression
 - Reducing Error (mse, mae)
 - Increasing r^2
 - Classification
 - Increasing accuracy score

Model Improvement Rivojlantrish yollari

1. Data Quality and Quantity
 - Koproq dataset collection
 - Datasetni tozalash
2. Feature Engineering
 - Yangi Featurelar yaratish
 - Domain knowledge
 - Transform features
 - Feature selections
3. Scaling
4. Turli xil algorithmlar bilan tekshirish
 - Bazi xollarda soddaroq algorithmlar ham yuqori natija beradi.
5. Hyper parameter tuning
 - Bizning algorithm va datasetga mos parametrlarni topish jarayoni
6. Cross Validation
7. Regularization
8. Ensemble Methods
9. Handling Imbalanced Dataset (Classification)

Yaxshi model $\Rightarrow$ Yaxshi dataset + Aqlli va foydali features + to'g'ri algorithmlar + tuning + overfittingni oldini olish

Transform Features

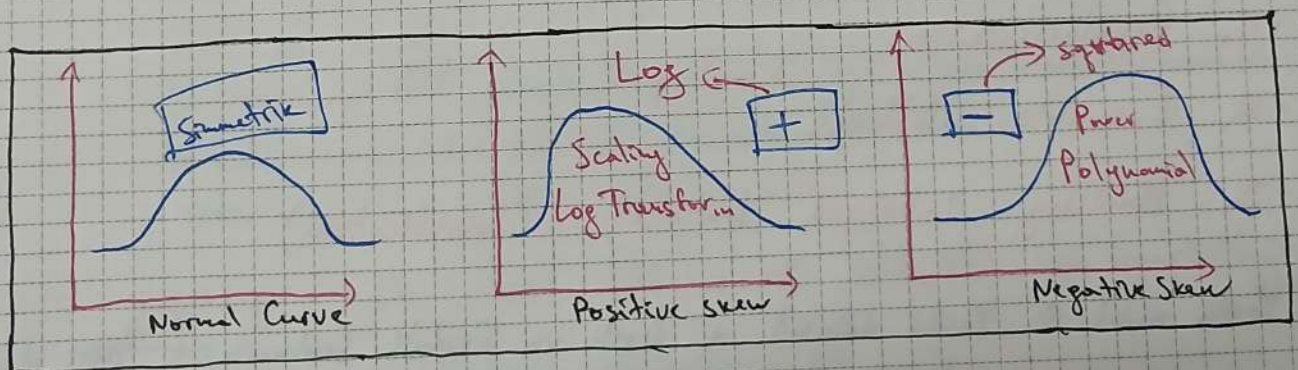
- Input featurelarni ML algoritmlar uchun moslashtirish jarayoni
- Yani featurelarni kompyuterga osongin tilda tushuntirish
- Qisqacha: Featurelarni o'zgartirish

Transform features: Turlari

- Feature Scaling
- Log Transformation ($\text{feature_log} = \log_{10}(1+x)$)
- Power Transformation (Qiymatlar past bo'lganda (x^2) oshirish)
- Polynomial Features (Bir haddan ko'p haddga o'tirish jarayoni $(x^2 + 2x + 3)$)
- Encoding

Transform features: Log Transformation

- Skewness (qiyshayishi)
- Matematik transformation
- Faqatgina musbat qiymatlarda ishlaydi (kattiq bolsa barcha qiymatlarga C-constant qiymat qo'shish)
- Asosan Positive skewnesslik holatlarda ishlaydi va musbat qiymatlarni normallashtirish
- Tree-based algoritmlarda bazi holatlarda foydasiz va keraksiz



Date.

No.

Features Transformation: Analyzing

```
m num_col = df.select_dtypes(include=['float64']) # Num_Col unique index
```

num_col

num_col.skew()

num_col zyghaygh dasgasini unique index

1. skew < 0.5

---> Transformation kerak emas

2. skew < 0

---> squared transformation

3. skew >= 0.5

---> Log transformation, sqrt()

import

numpy

as

np

log_transformation

= skewness [(skewness >= 0.5)].index.tolist()

Filter

log_transformation

logni qillish

for col in

log_transformation:

df[col] = np.log1p(df[col])